

# Lecture 1: Principles of Economics (Chapter 1)

*Applied Microeconomics — Master Notes · by Jakob V. Stangel*

**Required reading:** Friberg, ch. 1 (*Introduction*).

**Theme of the lecture (Butera):** the "economic way of thinking" — economics studies how people allocate scarce resources among competing needs, using **three principles** (optimization · equilibrium · empiricism) and **models** (simplified, testable) to understand and predict behaviour.

★ **The heart of the chapter:** economics = **doing the best you can with scarce resources**. The toolkit is **three principles** — **Optimization · Equilibrium · Empiricism**; the two workhorses of optimization are **opportunity cost** and **marginal thinking**; and the method is building **positive (testable) models**, not making normative judgments. "*All models are useful, but all models are wrong.*"

**The red thread.** Everything starts from **scarcity** — without it we wouldn't need economics. From scarcity flows: you must **choose** (optimize) → every choice has an opportunity cost and is decided **at the margin** → many people choosing interact until no one wants to move (**equilibrium**) → and we test our claims about all this with **models + data** (empiricism). For any decision ask: *what's scarce · what's the best alternative given up (opportunity cost) · what does one more unit give me (marginal) · is this claim positive (testable) or normative (a value judgment)?*

## Key concepts / "modes" to use

- **Scarcity** — limited resources vs unlimited wants; the reason economics exists.
- **The three principles: Optimization · Equilibrium · Empiricism.**
- Opportunity cost — the value of the *best alternative* you give up.
- Marginal thinking — decide by the value of *one more* unit; law of diminishing marginal returns.
- **Model** — a simplified representation of relationships between variables ("all models are wrong, but useful").
- **Positive vs normative** — a *testable* cause-effect claim vs a *value judgment*.

## Section-by-section main points

### What is economics?

**Core:** economics = **the study of how people allocate scarce resources over competing needs** — "people (try to) do the best they can for themselves with what they have."

- **Scarcity of two things:** the **things we want** (goods, services, jobs, education, relationships, recognition...) *and* the **means to get them** (money, **time**, **attention**, information, patience, willpower). Air to breathe isn't scarce → not an economic question.
- Every decision is shaped by **scarcity** (can't have it all), **other people** (you're not an island), and **the rules of the game** (you can't do whatever you want).

- **How societies allocate: central planners** (someone at the top decides who gets what — parents, government) vs **markets** (value set by supply & demand — from job markets to dating markets). Economics gives tools to predict how we decide and when markets do (or don't) allocate *efficiently*.

## The three principles of economics

**Core:** the whole course rests on three ideas.

| Principle             | Meaning                                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| ① <b>Optimization</b> | We choose the <b>best feasible option</b> given our limited resources (money, time, attention, willpower are all scarce).      |
| ② <b>Equilibrium</b>  | A situation in which <b>no one would benefit</b> from changing their <i>own</i> behaviour, given the choices of everyone else. |
| ③ <b>Empiricism</b>   | We build <b>models</b> → models generate <b>predictions</b> → we use <b>data/experiments</b> to test (falsify) them.           |

## Optimization — opportunity cost + marginal thinking

**Core:** to optimise, use two tools — opportunity cost (weigh each use against its best alternative — "apples to apples") and marginal thinking (what does *one more* unit give me?).

Opportunity cost = the value of the best alternative you give up — nothing "free" is free if it uses a scarce resource.

### OPPORTUNITY COST

$$OC(\text{action}) = \text{benefit}(\text{action}) - \text{value of the best alternative given up}$$

**Reads:** the true cost of a choice is what it gives you **minus** what you gave up to get it. — **OC(action)** = the opportunity cost of doing it • **benefit(action)** = the gain the action delivers • **value of the best alternative given up** = the payoff of the *next-best* option you sacrificed by choosing this one.

- **Social media:** 1 hr/day on TikTok isn't free — you give up sport, study, work. Monetise the time with the hourly wage (avg **118 DKK** in Denmark):

### YEARLY TIME-COST OF 1 HR/DAY ON TIKTOK

$$118 \text{ DKK/hr} \times 1 \text{ hr/day} \times 365 \text{ days} = \mathbf{43,070 \text{ DKK / year}}$$

**Reads:** price the time you *could* have been earning, added up over a year. — **118 DKK/hr** = Denmark's average hourly wage (i.e. the value of one hour of your time) • **1 hr/day** = the time you spend • **365 days** = a full year → **43,070 DKK** of wages forgone.

- **Trip to Berlin (worked example):** drive = 1,280 DKK & 14 h; fly = 1,930 DKK & 3 h. Flying frees **11 h** you could work:

### OPPORTUNITY COST OF DRIVING INSTEAD OF FLYING

$$OC(\text{drive}) = (1,930 - 1,280) - (118 \times 11) = 650 - 1,298 = -648 \text{ DKK} \Rightarrow \text{fly}$$

*Reads:* the money you save by driving, minus the value of the time driving wastes. —  $(1,930 - 1,280)$  = the extra *ticket* cost of flying (650 DKK) ·  $(118 \times 11)$  = the value of the **11 hours** flying saves you, priced at the 118 DKK/hr wage (1,298 DKK) · a **negative** result means driving is **648 DKK worse** than flying → **fly**.

*(Still a model — what if you enjoy the drive?)*

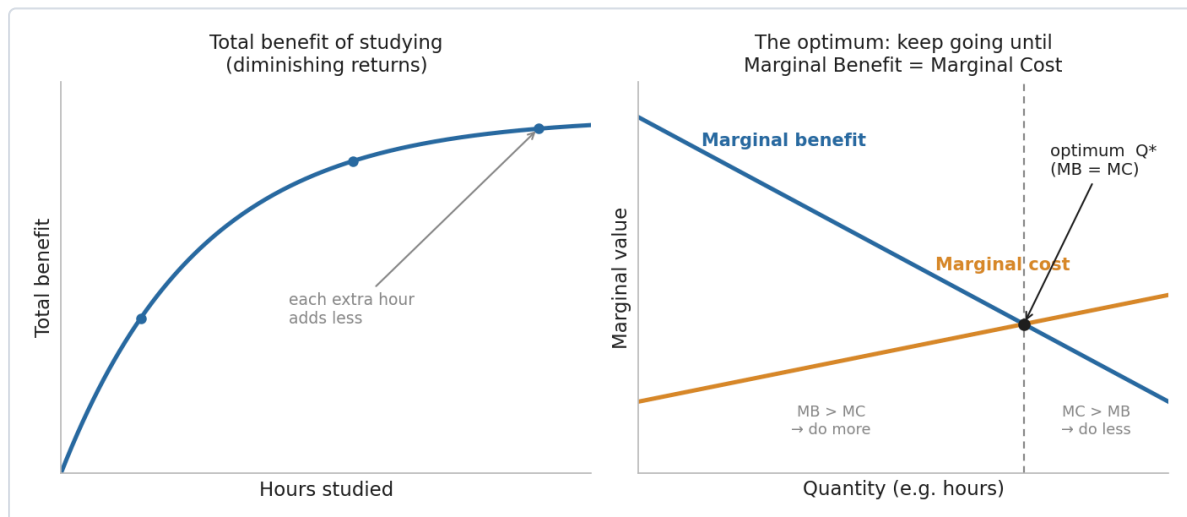
**Marginal thinking:** decide by the value of the **next** unit — keep going while marginal benefit exceeds marginal cost; stop where they meet. That point is the **optimum**:

### OPTIMUM CONDITION

choose the quantity where **Marginal Benefit = Marginal Cost** ( $MB = MC$ )

*Reads:* keep going until one more unit's benefit exactly equals its cost. — **Marginal Benefit (MB)** = the *extra* benefit from **one more** unit · **Marginal Cost (MC)** = the *extra* cost of that unit · at the **optimum**  $MB = MC$  (while  $MB > MC$  do more; once  $MC > MB$  you've gone too far).

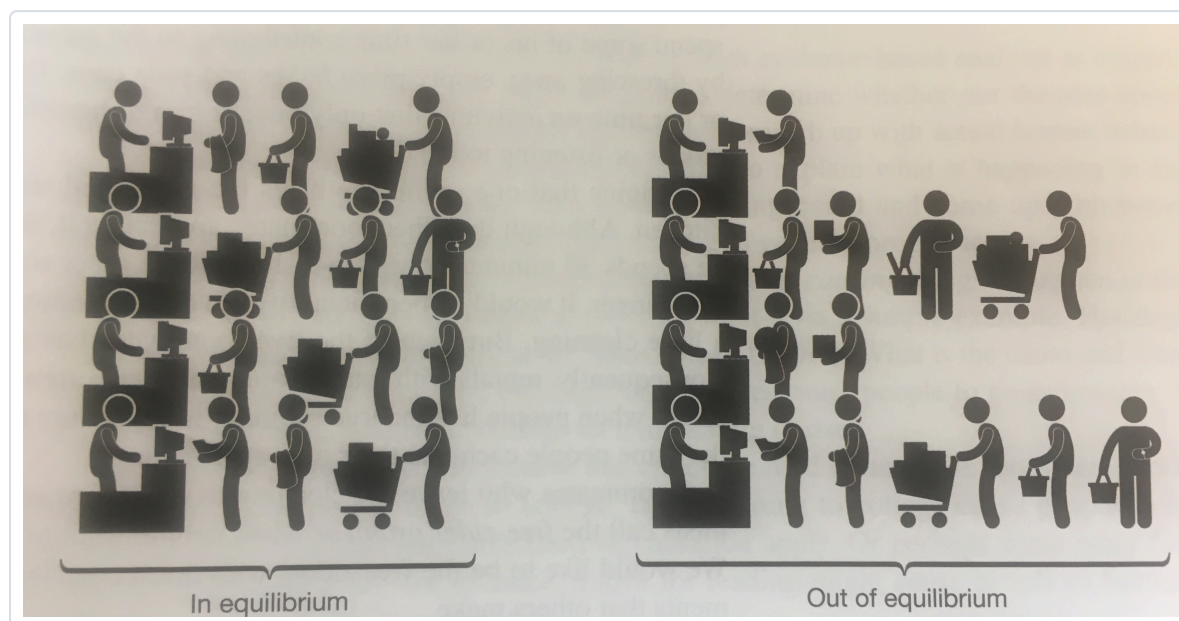
- *Night before the exam:* total benefit of studying rises then flattens; the **marginal benefit of each extra hour falls** (law of diminishing marginal returns). Optimal hours = where MB of one more hour just equals its MC.



**How to read it** (illustrative diagram — the graph the book draws in a later chapter). *Left:* total benefit rises but each extra hour adds less — the curve flattens (diminishing returns). *Right:* that flattening means **marginal benefit slopes down**; **marginal cost slopes up**. Left of the crossing,  $MB > MC$  so one more unit is worth it (*do more*); right of it,  $MC > MB$  (*do less*). The best you can do is the quantity where they meet — the **optimum  $Q^*$** , where  $MB = MC$ .

## Equilibrium

**Core:** an **equilibrium** is a resting point — given what everyone else is doing, **no individual can make themselves better off by changing their own behaviour**. Market prices are the classic example (later lectures): they adjust until quantity supplied = quantity demanded and no one wants to move.



**How to read it (equilibrium = supermarket queues, from the lecture).** Each lane is a checkout; the queue length is how many people are waiting. **Out of equilibrium** (right) the lanes are uneven, so anyone in a long line can still gain by switching to a shorter one — people keep moving. **In equilibrium** (left) all lanes are equal, so **no one can do better by changing lane on their own** — that "no incentive to move" resting point is exactly what equilibrium means (and why market prices settle where supply meets demand).

## Empiricism — models, and positive vs normative

**Core:** economics reasons with **models** and tests them against **data** — and it deals in **positive** (testable) statements, not **normative** (value) ones.

- A **model** expresses relationships between variables and makes **simplifying assumptions** (abstracts from reality). "*All models are useful, but all models are wrong*" — the art is choosing the right model (Keynes).



How to read it (Friberg Fig. 1.1 — what a "model" is). This is a Polynesian **stick chart**: sticks map the ocean's swell patterns, shells mark islands. It's not a literal picture of the Pacific — it **strips reality down to just the features you need to navigate**. That's the book's metaphor for an economic model: deliberately simplified, "wrong" as a full description, yet **good enough to steer decisions by**.

- **Positive vs normative:**
  - **Positive** = a **testable/falsifiable** cause-effect claim — *"each additional Starbucks in a ZIP code is associated with a 0.5% rise in housing prices"* (Glaeser et al. 2018).
  - **Normative** = a **value judgment** — *"gentrification is bad because it forces the poor out."*
  - **Economics is about positive statements**, not normative ones.
- **Correlation  $\neq$  causation**: in summer people eat more ice-cream *and* drown more — ice-cream doesn't cause drowning (a hidden cause: heat). A theory is only useful if it's **empirically falsifiable**.

- **Expanding models to fit data:** "people only give to charity because they care about others' well-being" fails the data (public spending doesn't crowd out giving 1-for-1 — Andreoni 1990; people give to *inefficient* charities). So we add "**joy of giving**" (utility from the act of giving itself).

## ▮ Friberg's framing — allocation mechanisms (Ch. 1's example)

**Core:** a scarce good (e.g. **university places**) can be allocated many ways — and each has different effects on **efficiency** and **equity**. This is the book's opening illustration of "the role of economics."

- **Non-price mechanisms:** aptitude tests/interviews · secondary-school grades · open entry + hard exams · **waiting lists** · **queues** ("camp on the sidewalk") · **lotteries** · social class/legacy.
- **Price mechanisms:** everyone pays the **same fee** (set high enough to clear demand) · **auction** places to the highest bidders.
- Real systems **combine** several (fees + tests + interviews). Micro gives the tools to analyse the **trade-offs** (who gets in, how tied to parents' income, effect on effort).

## Cases & examples

Each: *the example* → *the concept it teaches*.

- **1 hr/day of TikTok** → opportunity cost ( $\approx 43,070$  DKK/yr at the average wage) — nothing using scarce time is free.
- **Drive vs fly to Berlin** → opportunity cost done properly: include the *value of time*, and driving turns out to *cost* 648 DKK more.
- **Studying the night before an exam** → marginal thinking + **diminishing returns**; optimise where  $MB = MC$ .
- **Ice-cream & drowning** → **correlation  $\neq$  causation**.
- **Starbucks & house prices** (Glaeser 2018) → a **positive** statement; **gentrification "is bad"** → a **normative** one.
- **Charitable giving** (Andreoni 1990) → models must be **expanded to fit data** ("joy of giving").
- **University admissions** (Friberg) → **allocation mechanisms** (price vs non-price) and their equity/efficiency trade-offs.

## Key terms

| Term                                | Meaning                                                           |
|-------------------------------------|-------------------------------------------------------------------|
| Scarcity                            | Limited resources vs unlimited wants — the basis of economics     |
| Optimization                        | Choosing the best feasible option given constraints               |
| Equilibrium                         | No one gains from changing their own behaviour, given others'     |
| Empiricism                          | Build models → predict → test with data                           |
| Opportunity cost                    | Value of the best alternative given up                            |
| Marginal thinking                   | Deciding by the value of one more unit                            |
| Law of diminishing marginal returns | Each extra unit adds less than the last                           |
| Model                               | Simplified representation of relationships between variables      |
| Positive statement                  | A testable/falsifiable cause-effect claim                         |
| Normative statement                 | A value judgment (good/bad, should/ought)                         |
| Allocation mechanism                | Rule that decides who gets a scarce good (price/lottery/queue...) |

## Exam pointers

- State the **three principles** (Optimization · Equilibrium · Empiricism) and define each.
- **Compute an opportunity cost** properly (include the value of time) — the Berlin example is the model answer.
- Use **marginal thinking** +  $MB = MC$  to find an optimum, and name the **law of diminishing marginal returns**.
- Classify a statement as **positive vs normative**, and remember **economics = positive**; watch for **correlation  $\neq$  causation**.
- Explain why economists use **models** ("all models are wrong, but useful") and give an **allocation-mechanism** trade-off (Friberg's university example).

## ★ Fun facts & memorable details

Sticky lines from Lecture 1.

- Keynes: "*Economics is a science of thinking in terms of models joined to the art of choosing models which are relevant to the contemporary world.*"
- The course mantra: "*All models are useful, but all models are wrong.*"
- **1 hour of TikTok a day  $\approx$  43,070 DKK a year** in forgone wages (at Denmark's  $\sim$ 118 DKK average hourly wage).

- Driving to Berlin *looks* cheaper (1,280 vs 1,930 DKK) but, once you price the 11 lost hours, it actually **costs 648 DKK more** than flying.
  - "*During the summer people eat more ice-cream and also drown more*" — the classic **correlation-≠-causation** trap (the hidden cause is heat).
  - "**Joy of giving**" (Andreoni 1990): we don't donate only for others' welfare — we get utility from the *act* of giving, which is why public spending doesn't crowd out private giving 1-for-1.
  - Learning micro is "*a first course in a new language*" — a bit hard, not immediately intuitive, needs practice.
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